Portfolio Optimization Project Documentation

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1. Project Overview

What This Project Does

This portfolio optimization tool implements **Modern Portfolio Theory (MPT)**, developed by Harry Markowitz in 1952 (he won a Nobel Prize for it). The project:

- Downloads historical stock price data
- Calculates expected returns and risks for individual stocks
- Finds the optimal portfolio allocation that maximizes risk-adjusted returns
- Generates the efficient frontier showing all optimal portfolios
- Creates visualizations comparing different investment strategies

Why Portfolio Optimization Matters

Imagine you have $10,000 to invest across multiple stocks. How much should you put in each? Portfolio optimization answers this by:

- **Maximizing returns** for a given level of risk
- **Minimizing risk** for a given target return
- **Exploiting diversification** - the only "free lunch" in finance

The key insight: By combining stocks that don't move perfectly together, you can reduce risk without sacrificing returns.

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2. Mathematical Foundations

2.1 Returns

Daily Return Formula:

$Return_t = (Price_t - Price_{t-1}) / Price_{t-1}$

Example: Stock goes from $100 to $102

Return = (102 - 100) / 100 = 0.02 = 2%

Annualization:

Annual Return = Mean Daily Return × 252

We use 252 because there are approximately 252 trading days per year (365 days minus weekends and holidays).

Why use returns instead of prices?

- Returns are comparable across stocks (a $5 move means different things for a $10 stock vs $500 stock)
- Returns are more statistically stable
- We can aggregate returns across time periods

2.2 Risk (Volatility)

Standard Deviation Formula:

$\sigma = \mathrm{sqrt}(\Sigma(Return_i - Mean\_Return)^2 / n)$

This measures how much returns bounce around their average. Higher volatility = more uncertainty = more risk.

Annualization:

Annual Volatility = Daily Std Dev × sqrt(252)

Why square root? Because variance scales linearly with time, and standard deviation is the square root of variance.

Example:

- Stock A: Returns are consistently 0.5% to 1.5% daily → Low volatility (predictable)
- Stock B: Returns swing from -5% to +8% daily → High volatility (unpredictable)

2.3 Correlation and Covariance

Correlation measures how two stocks move together:

- **+1**: Perfect positive correlation (always move together)
- **0**: No relationship (independent)
- **-1**: Perfect negative correlation (move in opposite directions)

Covariance is the unstandardized version of correlation:

Cov(A, B) = Average[(Return_A - Mean_A) × (Return_B - Mean_B)]

Why it matters:

- If Apple and Microsoft both drop when tech has bad news, they're positively correlated
- Diversification works best with low or negative correlations
- The covariance matrix captures all pairwise relationships

Covariance Matrix Example (3 stocks):

	AAPL	MSFT	JNJ
AAPL	0.058	0.042	0.018
MSFT	0.042	0.051	0.016
JNJ	0.018	0.016	0.032

- Diagonal: Each stock's variance (risk with itself)
- Off-diagonal: Covariance between pairs
- Symmetric: Cov(A,B) = Cov(B,A)

2.4 Portfolio Return

Formula:

Portfolio Return = $w_1R_1 + w_2R_2 + ... + w_nR_n$

Or in vector notation:

$R_p = w^T \cdot \mu$

Where:

- **w** = vector of weights $[w_1, w_2, ..., w_n]$
- **μ** = vector of expected returns $[R_1, R_2, ..., R_n]$

Example: Portfolio with 3 stocks:

- 50% in AAPL (30% expected return)
- 30% in JNJ (10% expected return)
- 20% in XOM (20% expected return)

Portfolio Return = 0.5×30% + 0.3×10% + 0.2×20%

= 15% + 3% + 4% = 22%

Key insight: Portfolio return is just a weighted average. Simple!

2.5 Portfolio Risk (The Complex Part)

Formula:

Portfolio Variance = $w^T \cdot \Sigma \cdot w$

Portfolio Volatility = sqrt(Portfolio Variance)

Where:

- **w** = weights vector
- **Σ** = covariance matrix
- **w^T** = transposed weights

Why it's NOT a simple average:

Example with 2 stocks, each with 20% volatility:

If perfectly correlated (correlation = 1):

- 50% in each stock
- Portfolio volatility = 20% (no diversification benefit)

If uncorrelated (correlation = 0):

- 50% in each stock
- Portfolio volatility = 14.1% (diversification magic!)

If negatively correlated (correlation = -1):

- 50% in each stock
- Portfolio volatility ≈ 0% (perfect hedge)

The covariance matrix captures these relationships automatically.

Expanded formula for 2 stocks:

$\sigma\_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2Cov(1,2)$

That last term ($2w_1w_2Cov$) is crucial - it's the diversification effect!

2.6 Sharpe Ratio

Formula:

Sharpe Ratio = (Portfolio Return - Risk-Free Rate) / Portfolio Volatility

What it means: Return per unit of risk. Higher is better.

Example:

- Portfolio A: 15% return, 20% volatility, risk-free rate = 2%
    - Sharpe = (15% - 2%) / 20% = 0.65
- Portfolio B: 10% return, 5% volatility, risk-free rate = 2%
    - Sharpe = (10% - 2%) / 5% = 1.60

Portfolio B is better! Even though it has lower returns, it's much more efficient.

Interpretation:

- Sharpe < 1: Mediocre risk-adjusted returns
- Sharpe 1-2: Good performance
- Sharpe > 2: Excellent (rare in real markets)
- Sharpe > 3: Suspicious (check your data!)

Risk-free rate: We use ~2% representing US Treasury bonds - the "safe" investment baseline.

2.7 The Efficient Frontier

Definition: The set of portfolios that offer the maximum return for a given level of risk, or minimum risk for a given return.

Key insight: Any portfolio NOT on the frontier is suboptimal. You could either:

- Get the same return with less risk, OR
- Get higher return with the same risk

Shape: The frontier curves upward - at low risk levels, you get big return increases for small risk increases. At high risk levels, returns plateau (diminishing returns).

Mathematical formulation:

For each target return R^*, solve:

Minimize: $w^T \cdot \Sigma \cdot w$ (portfolio variance)

Subject to:

- $w^T \cdot \mu = R^*$ (achieve target return)

- $\Sigma w_i = 1$ (weights sum to 100%)

- $w_i \geq 0$ (no short selling)

Run this for many target returns, plot them all → you get the efficient frontier curve.

2.8 The Optimization Problem

What we're solving:

Find weights **w** that maximize Sharpe ratio:

Maximize: $(w^T \cdot \mu - r_f) / \text{sqrt}(w^T \cdot \Sigma \cdot w)$

Constraints:

1. **$\Sigma w_i = 1$** (invest 100% of capital)
2. **$w_i \geq 0$** (long-only, no short selling)
3. **Optional: $w_i \leq max\_weight$** (limit concentration)
4. **Optional: $w_i \geq min\_weight$** (force diversification)

Why it's hard: This is a non-linear constrained optimization problem. We can't just set derivatives to zero - we need numerical algorithms.

Solution method: SLSQP (Sequential Least Squares Programming) - an algorithm that:

- Starts with an initial guess (equal weights)
- Iteratively improves the solution
- Respects all constraints
- Converges to the optimal portfolio

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3. Code Structure and Flow

3.1 Configuration Section

tickers = ['AAPL', 'JPM', 'JNJ', 'INTC', 'YM=F']

risk_free_rate = 0.02

MAX_WEIGHT = 0.40 # No more than 40% in any stock

MIN_WEIGHT = 0.05 # At least 5% in each stock

USE_MAX_CONSTRAINT = True

USE_MIN_CONSTRAINT = False

What it does:

- Define which stocks to analyze
- Set the risk-free rate (US Treasury rate)
- Configure portfolio constraints

Why constraints matter:

- Without constraints, optimizer might put 100% in best stock (concentration risk)
- Real portfolio managers always limit single positions
- Constraints make results more realistic and practical

3.2 Phase 1: Data Download and Preparation

```
data = yf.download(tickers, start='2020-01-01', end='2024-01-01',
                   progress=False, auto_adjust=False)

data = data['Adj Close']

data = data.dropna()

returns = data.pct_change().dropna()
```

Step-by-step:

1. **Download data** from Yahoo Finance
   - `auto_adjust=False` keeps 'Adj Close' column
   - Date range: 2020-2024 (4 years of data)
2. **Extract Adjusted Close prices**
   - Why 'Adj Close'? Accounts for stock splits and dividends
   - Example: Apple did 4-for-1 split in 2020. Adj Close adjusts historical prices so there's no fake -75% drop
3. **Remove missing data**
   - Some stocks might have missing days (holidays, trading halts)
   - We need complete data for all stocks on same dates
4. **Calculate returns**
   - `pct_change()` computes (price_today - price_yesterday) / price_yesterday
   - `dropna()` removes first row (no previous day to compare)

Output: DataFrame with daily returns for each stock

3.3 Phase 2: Calculate Statistics

```
mean_returns = returns.mean() * 252

cov_matrix = returns.cov() * 252
```

```
correlation = returns.corr()
```

What's calculated:

1. **Mean returns** (annualized)
   - Average daily return × 252 trading days
   - This is our "expected return" estimate
2. **Covariance matrix** (annualized)
   - Shows how each pair of stocks moves together
   - Multiply by 252 to annualize (variance scales linearly)
3. **Correlation matrix**
   - Standardized version of covariance
   - Easier to interpret (values between -1 and +1)

Key assumption: Historical averages will continue into the future. This is a major limitation!

3.4 Phase 3: Portfolio Optimization

Core Functions

portfolio_stats():

```
def portfolio_stats(weights, mean_returns, cov_matrix, risk_free_rate):

    portfolio_return = np.dot(weights, mean_returns)

    portfolio_variance = np.dot(weights, np.dot(cov_matrix, weights))

    portfolio_volatility = np.sqrt(portfolio_variance)

    sharpe_ratio = (portfolio_return - risk_free_rate) / portfolio_volatility

    return portfolio_return, portfolio_volatility, sharpe_ratio
```

What it does: Given a set of weights, calculate portfolio metrics

- **Return:** Weighted average of stock returns
- **Variance:** Accounts for covariances (the w^T·Σ·w calculation)

- **Volatility:** Square root of variance
- **Sharpe:** Risk-adjusted return

negative_sharpe():

```
def negative_sharpe(weights, mean_returns, cov_matrix, risk_free_rate):
    return -portfolio_stats(weights, mean_returns, cov_matrix, risk_free_rate)[2]
```

Why negative? The optimizer minimizes functions, but we want to maximize Sharpe. Solution: minimize negative Sharpe (same thing as maximizing Sharpe).

optimize_sharpe_with_constraints():

```
def optimize_sharpe_with_constraints(mean_returns, cov_matrix, risk_free_rate,
                                     max_weight=None, min_weight=None):
    num_assets = len(mean_returns)
    initial_weights = np.array([1/num_assets] * num_assets)

    constraints = [{'type': 'eq', 'fun': lambda w: np.sum(w) - 1}]

    if max_weight is not None:
        bounds = tuple((0, max_weight) for _ in range(num_assets))
    else:
        bounds = tuple((0, 1) for _ in range(num_assets))

    result = minimize(negative_sharpe, initial_weights,
                 args=(mean_returns, cov_matrix, risk_free_rate),
                 method='SLSQP', bounds=bounds, constraints=constraints)
```

    return result

How it works:

1. **Initial guess:** Start with equal weights (20% each for 5 stocks)

2. **Constraints:**

    - `{'type': 'eq', 'fun': lambda w: np.sum(w) - 1}` means "this function must equal zero"
    - So np.sum(w) - 1 = 0, which means np.sum(w) = 1 (weights sum to 100%)

3. **Bounds:**

    - Each weight must be between 0 and max_weight (e.g., 0 to 0.40)
    - Prevents short selling (negative weights)
    - Prevents over-concentration

4. **Optimization:**

    - SLSQP algorithm iteratively adjusts weights
    - Tries to minimize negative Sharpe (maximize Sharpe)
    - Respects all constraints
    - Returns optimal weights

optimize_for_return():

```
def optimize_for_return(target_return, mean_returns, cov_matrix,

                       max_weight=None, min_weight=None):

    # Similar to above, but with additional constraint:

    constraints = [

        {'type': 'eq', 'fun': lambda w: np.sum(w) - 1},

        {'type': 'eq', 'fun': lambda w: np.dot(w, mean_returns) - target_return}

    ]
```

```
    # Minimize variance (not negative Sharpe)

    result = minimize(lambda w: np.dot(w, np.dot(cov_matrix, w)),

                     initial_weights, method='SLSQP',

                     bounds=bounds, constraints=constraints)

    return result
```

What it does: Find minimum risk portfolio for a specific target return

- Used to generate the efficient frontier
- Run this for many target returns (e.g., 5%, 10%, 15%, ..., 30%)
- Plot all results → efficient frontier curve

3.5 Phase 4: Generate Efficient Frontier

```
min_return = mean_returns.min()

max_return = mean_returns.max()

target_returns = np.linspace(min_return, max_return, 50)

efficient_portfolios = []

for target in target_returns:

    result = optimize_for_return(target, mean_returns, cov_matrix,

                                max_weight=max_w, min_weight=min_w)

    if result.success:

        # Store portfolio stats

        efficient_portfolios.append({...})
```

Process:

1. Find range of possible returns (min to max of individual stocks)
2. Create 50 evenly spaced target returns
3. For each target, find the minimum-risk portfolio that achieves it
4. Store all successful optimizations

Result: A curve showing the best possible risk-return tradeoffs

3.6 Phase 5: Visualizations

The code generates 6 key visualizations (explained in detail in section 5).

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4. Understanding the Outputs

4.1 Console Output

Individual Asset Statistics:

	Annual Return	Annual Volatility	Sharpe Ratio
AAPL	29.90	33.57	0.831
INTC	6.57	41.26	0.111
JNJ	6.60	20.64	0.223
JPM	13.62	34.42	0.338
YM=F	9.41	22.23	0.333

What this tells you:

- **AAPL:** High return (30%), high risk (34%), excellent Sharpe (0.83)
- **INTC:** Low return (7%), very high risk (41%), terrible Sharpe (0.11)
    - Intel had a bad period - high risk, low reward
- **JNJ:** Low return (7%), low risk (21%), mediocre Sharpe (0.22)
    - Defensive stock - stable but boring
- **JPM & YM=F:** Moderate risk/return

Optimal Portfolio:

Optimal Weights:

AAPL : 46.10%

MSFT : 31.90%

XOM   : 22.10%

JNJ   : 0.00%

JPM   : 0.00%

Expected Annual Return: 27.00%

Annual Volatility (Risk): 28.00%

Sharpe Ratio: 0.893

Interpretation:

- Optimizer chose 3 stocks, excluded 2
- Heavy weight in AAPL (best Sharpe ratio)
- Significant MSFT and XOM for diversification
- Zero weight in JNJ and JPM (don't improve risk-adjusted returns)
- Portfolio Sharpe (0.893) beats all individual stocks except AAPL

Equal-Weight Portfolio:

Expected Annual Return: 18.00%

Annual Volatility (Risk): 30.00%

Sharpe Ratio: 0.533

Comparison:

- Optimal: 27% return, 28% risk, 0.893 Sharpe
- Equal-weight: 18% return, 30% risk, 0.533 Sharpe

Conclusion: Optimization adds significant value!

- 9% higher return
- 2% lower risk
- 67% higher Sharpe ratio

4.2 CSV Output Files

optimal_portfolio_weights.csv:

Ticker,Weight,Weight_Percent

AAPL,0.461,46.1

MSFT,0.319,31.9

XOM,0.221,22.1

JNJ,0.000,0.0

JPM,0.000,0.0

Use case: Import into spreadsheet to track actual portfolio

efficient_frontier.csv:

return,volatility,sharpe,weights

0.0657,0.2064,0.223,[0,1,0,0,0]

0.0800,0.2100,0.286,[0.1,0.8,0.1,0,0]

...

0.2990,0.3357,0.831,[1,0,0,0,0]

Use case: Analyze tradeoffs, find portfolios with specific risk levels

individual_assets.csv: All individual stock statistics for further analysis

correlation_matrix.csv: Full correlation matrix - useful for understanding relationships

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5. Interpreting the Visualizations

5.1 Correlation Matrix Heatmap

What it shows: How stocks move together

Reading the chart:

- **Red (positive):** Stocks move in same direction
  - Example: AAPL-MSFT correlation of 0.78 means when Apple rises, Microsoft usually rises too
- **Blue (negative):** Stocks move in opposite directions
  - Rare in practice - most stocks have positive correlation
- **White (zero):** No relationship
- **Diagonal is always 1.0:** Stock perfectly correlated with itself

Key insights to look for:

1. **Tech stocks cluster:** AAPL, MSFT, GOOGL, NVDA often show 0.6-0.8 correlation
   - They respond to same news (interest rates, tech regulations, economic cycles)
   - Less diversification benefit within tech
2. **Cross-sector diversification:** Tech vs Consumer Staples might show 0.3-0.5
   - Better diversification - they don't always move together
3. **Defensive stocks:** JNJ, PG, KO often have lower correlations with everything
   - "Defensive" means they're more stable, less affected by market swings

Practical application:

- High correlation = less diversification benefit
- Low correlation = better risk reduction through diversification
- Negative correlation = holy grail (rare and valuable)

5.2 Normalized Price History

What it shows: How $100 invested in each stock would have grown

Why normalize to 100?

- Makes performance comparable regardless of actual price
- $10 stock going to $20 = same as $1000 stock going to $2000 (both 100→200)

Reading the chart:

Example interpretation:

- AAPL: 100 → 270 (170% gain)
- JNJ: 100 → 120 (20% gain)
- INTC: 100 → 95 (-5% loss)

Key patterns to notice:

1. **Volatility visible in wiggles:**
    - Smooth line = low volatility (JNJ)
    - Jagged line = high volatility (tech stocks)
2. **Correlation visible in co-movement:**
    - Lines moving together = high correlation
    - Lines diverging = low correlation
3. **Drawdowns (drops from peaks):**
    - How much did stock fall from its highest point?
    - Important risk measure not captured by standard deviation
4. **Recovery patterns:**
    - Some stocks quickly recover from drops (resilient)
    - Others stay depressed (risky)

Practical insight: Past performance ≠ future results, but this shows:

- Which stocks were winners/losers in this period
- How stable or volatile they were
- Whether they moved together or independently

5.3 Efficient Frontier Plot

The most important chart!

Components:

1. **Blue curve (Efficient Frontier):**

    - Every point = an optimal portfolio
    - Left side: Low risk, low return (mostly defensive stocks)
    - Right side: High risk, high return (mostly growth stocks)
    - **Key property:** No portfolio above this curve is achievable

2. **Red dots (Individual Assets):**

    - Each stock plotted at its risk-return point
    - Usually scattered around, some on frontier, some below
    - Shows you can't beat diversification

3. **Gold star (Optimal Portfolio - Max Sharpe):**

    - The single best risk-adjusted portfolio
    - Usually in the middle of the frontier
    - This is THE answer to "what should I invest in?"

4. **Green diamond (Equal-Weight Portfolio):**

    - Benchmark: what you'd get with naive diversification
    - Usually below the frontier (suboptimal)
    - Shows value of optimization

Reading the chart:

Scenario 1: Optimal dominates individual stocks

- Optimal portfolio: 25% return, 20% risk
- Best individual stock: 30% return, 35% risk
- **Conclusion:** Diversification reduced risk more than it reduced return

Scenario 2: One stock dominates (problem!)

- If optimal portfolio sits right on top of one stock (like AAPL)
- Means that stock had such good Sharpe ratio, optimizer put 100% in it
- **Solution:** Add max weight constraints

Scenario 3: Equal-weight vs Optimal

- If equal-weight is far below frontier: optimization adds significant value
- If equal-weight is close to frontier: stocks were already well-diversified

The curve's shape:

- **Steep left side:** Small risk increases give big return gains (efficient region)
- **Flat right side:** Big risk increases give small return gains (inefficient region)
- **Implication:** Don't chase maximum returns - the risk-return tradeoff gets worse

Practical use:

- Risk-averse investor: Choose portfolio on left side of frontier
- Risk-tolerant investor: Choose portfolio on right side
- Rational investor: Choose the gold star (max Sharpe)

5.4 Portfolio Allocation Pie Chart

What it shows: How to split your money in the optimal portfolio

Example interpretation:

AAPL: 46.1% → Invest $4,610 of every $10,000

MSFT: 31.9% → Invest $3,190

XOM: 22.1% → Invest $2,210

What to look for:

1. **Concentration:**

    - Is one stock >50%? High concentration risk
    - More evenly distributed? Better diversification

2. **Sector exposure:**

    - All tech? Sector concentration risk
    - Mixed sectors? Better diversification

3. **Excluded stocks (0% weight):**

    - Why excluded? Usually poor Sharpe ratio

    - They don't improve risk-adjusted returns

Common patterns:

Without constraints:

- Might see 80-100% in one or two stocks
- Mathematically optimal but practically risky

With MAX_WEIGHT = 0.40:

- More balanced: top stock capped at 40%
- Forces diversification
- More realistic for actual investing

With MIN_WEIGHT = 0.05:

- Every stock gets at least 5%
- Very diversified
- May sacrifice some Sharpe ratio for safety

Practical application: If you have $10,000 to invest and the pie chart shows:

- 46% AAPL → Buy $4,600 worth of Apple stock
- 32% MSFT → Buy $3,200 worth of Microsoft
- 22% XOM → Buy $2,200 worth of Exxon

5.5 Risk-Return Comparison Bar Charts

Two side-by-side charts:

Left: Expected Returns

- Shows annual return for each stock + portfolios
- Higher bars = higher expected returns
- Optimal portfolio usually competitive with best stocks

Right: Annual Volatility (Risk)

- Shows risk for each stock + portfolios
- Higher bars = more volatile (risky)
- **Key insight:** Optimal portfolio often has LOWER risk than individual stocks

What to look for:

1. **Return-risk tradeoff:**

    - High return stocks often have high risk (AAPL, NVDA)
    - Low return stocks often have low risk (JNJ, KO)
    - Is the extra return worth the extra risk?

2. **Optimal vs Equal-Weight:**

    - **Ideal:** Optimal has higher return AND lower risk
    - **Common:** Optimal has higher return, similar risk
    - **Concerning:** If equal-weight beats optimal, something's wrong

3. **Diversification benefit visible:**

    - If optimal portfolio has lower risk than weighted average of individual stocks
    - This is diversification at work!

Example interpretation:

Stock A: 30% return, 40% risk Stock B: 10% return, 15% risk Optimal Portfolio (50-50): 20% return, 22% risk

Notice: Portfolio risk (22%) < average of individual risks (27.5%) This is the magic of diversification!

5.6 Sharpe Ratio Comparison

What it shows: Risk-adjusted returns - the ultimate metric

Reading the chart:

- Higher bars = better risk-adjusted performance
- Green dashed line at 1.0 = benchmark for "good" performance

Typical patterns:

1. **Optimal portfolio has highest Sharpe:**

    - That's the goal! Maximum risk-adjusted return
    - If it doesn't, optimization failed or constraints too restrictive

2. **Wide variation in individual Sharpes:**
    - Some stocks very efficient (high Sharpe)
    - Others inefficient (low Sharpe)
    - Explains why optimizer excludes some stocks
3. **Equal-weight usually lower than optimal:**
    - Shows value of optimization
    - Gap = how much optimization improved risk-adjusted returns

Example interpretation:

AAPL: 0.83 → For every 1% of risk, get 0.83% excess return

INTC: 0.11 → For every 1% of risk, get only 0.11% excess return

Optimal: 0.89 → Best risk-adjusted returns

Equal: 0.53 → Naive diversification

Conclusion:

- AAPL is efficient, INTC is terrible
- Optimization improved Sharpe by 68% vs equal-weight
- This justifies the complexity of optimization

Practical meaning:

- Sharpe 0.5: Mediocre - you're not being compensated well for risk
- Sharpe 1.0: Good - decent risk-adjusted returns
- Sharpe 1.5+: Excellent - very efficient portfolio
- Sharpe 2.0+: Exceptional (rare in real markets)

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6. Practical Applications

6.1 How to Use This Tool

Step 1: Choose your stocks

tickers = ['AAPL', 'MSFT', 'GOOGL', 'JNJ', 'PG', 'JPM', 'XOM', 'DIS', 'NVDA', 'KO']

Considerations:

- Mix sectors (tech, healthcare, finance, energy, consumer)
- Include some defensive stocks (JNJ, PG, KO)
- Include some growth stocks (AAPL, NVDA, GOOGL)
- 8-15 stocks is a good range (enough diversification, not too complex)

Step 2: Set constraints

MAX_WEIGHT = 0.30 # No more than 30% in any stock

MIN_WEIGHT = 0.05 # At least 5% in each stock

Guidelines:

- **Conservative:** MAX = 0.25, MIN = 0.10 (very diversified)
- **Moderate:** MAX = 0.40, MIN = 0.05 (balanced)
- **Aggressive:** MAX = 0.60, MIN = None (allow concentration)

Step 3: Run the optimization

Step 4: Interpret results

- Look at optimal weights - do they make sense?
- Check Sharpe ratio - is it significantly better than equal-weight?
- Review efficient frontier - is optimal portfolio on the curve?

Step 5: Implement in real portfolio

- Use the weight percentages to allocate your capital
- Example: $10,000 × 46% = $4,600 in AAPL

6.2 Rebalancing

The problem: Over time, weights drift from optimal

Example:

- Start: 50% AAPL, 50% JNJ
- AAPL doubles, JNJ stays flat
- Now: 67% AAPL, 33% JNJ (drifted from target)

Solution: Periodic rebalancing

How often:

- **Quarterly:** Common for active investors
- **Annually:** Sufficient for most people
- **Threshold-based:** Rebalance when any weight drifts >5% from target

Process:

1. Re-run optimization with updated data
2. Compare current weights to new optimal weights
3. Buy/sell to realign

Costs to consider:

- Trading commissions (usually minimal now)
- Bid-ask spreads
- Tax implications (capital gains)

6.3 Limitations and Reality Checks

This tool assumes:

1. **Past returns predict future returns** (often wrong!)
2. **Returns follow normal distribution** (fat tails in reality)
3. **Correlations stay constant** (they change in crises)
4. **No transaction costs** (real trading has costs)
5. **Infinite liquidity** (can buy/sell any amount instantly)
6. **No taxes** (capital gains matter!)

Real-world adjustments:

1. Don't trust historical returns blindly

- If AAPL returned 30% annually 2020-2024, don't assume 30% forever
- Consider using more conservative estimates
- Maybe adjust to long-term market average (~10%)

2. Add practical constraints

- Minimum position size (hard to manage 50 tiny positions)

- Transaction cost thresholds (don't trade for <1% improvement)
- Tax-loss harvesting opportunities

3. Consider factors not in the model

- Your personal risk tolerance
- Investment time horizon
- Liquidity needs
- Tax situation
- Existing holdings (don't double-count)

4. Stress test

- What if tech crashes 50%? (Your portfolio heavily tech?)
- What if interest rates spike? (Affects different sectors differently)
- What if we enter recession? (Defensive stocks outperform)

6.4 Extensions and Improvements

Things you could add:

1. Multiple time periods

- Compare 2015-2020 vs 2020-2024
- Are optimal portfolios stable or wildly different?
- Use rolling windows for robustness

2. Monte Carlo simulation

- Generate thousands of random portfolios
- Plot them all on risk-return space
- Shows how special the optimal portfolio is

3. Backtesting

- Use 2015-2020 data to optimize
- Test that portfolio on 2020-2024 data
- Does optimization actually work out-of-sample?

4. Factor models

- Instead of just historical returns, use factor models (Fama-French)
- More sophisticated estimate of expected returns

- Better theoretical foundation

5. Black-Litterman model

- Combine market equilibrium with your personal views
- More stable, less extreme portfolios
- Used by professional investors

6. Risk parity

- Instead of maximizing Sharpe, equalize risk contribution
- Each stock contributes equally to portfolio risk
- Often more diversified

7. Minimum variance portfolio

- Ignore returns, just minimize risk
- Often surprisingly good performance
- Less sensitive to return estimation errors

8. Transaction cost modeling

- Add explicit costs for buying/selling
- Optimization balances improvement vs costs
- More realistic for frequent rebalancing

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7. Limitations and Considerations

7.1 The Fundamental Problem: Garbage In, Garbage Out

The model is only as good as the inputs:

Expected returns are notoriously hard to estimate:

- Historical average might be 30% (AAPL 2020-2024)
- But true expected future return might be 10%
- Small errors in return estimates → big errors in optimal weights

Example of sensitivity:

- If you overestimate AAPL return by 5%, optimizer puts 80% in AAPL
- If you underestimate, optimizer excludes AAPL entirely
- Results are extremely sensitive to return estimates

Covariances are more stable but still change:

- Correlations spike toward 1.0 during market crashes ("everything falls together")
- Your diversification benefit disappears exactly when you need it most

7.2 Statistical Assumptions

The model assumes returns are normally distributed:

Reality:

- Returns have "fat tails" (extreme events more common than normal distribution predicts)
- Black Monday 1987: -22% in one day (should be impossible under normal distribution)
- 2008 financial crisis: correlations went to 1.0 (diversification failed)

Implications:

- Model underestimates extreme risk
- Actual worst-case scenarios worse than model predicts
- Need stress testing beyond the model

7.3 The Overfitting Problem

Example scenario:

You optimize on 2020-2024 data:

- Optimal portfolio: 80% NVDA, 20% AAPL
- Sharpe ratio: 1.5 (excellent!)

But 2020-2024 was exceptional for tech:

- AI boom drove NVDA up 10x
- Pandemic accelerated digital transformation
- Low interest rates favored growth stocks

In 2025-2029:

- Interest rates stay high (hurts growth stocks)
- AI hype fades (NVDA stagnates)

- Energy outperforms (but you have 0% energy)
- Your "optimal" portfolio underperforms

The problem: You optimized for the past, not the future.

Solutions:

- Use longer time periods (10+ years)
- Use multiple periods and average
- Add constraints to prevent extreme allocations
- Be skeptical of portfolios that seem "too good"

7.4 Practical Constraints Not in Model

Things real investors face:

1. Minimum investment amounts:

- Can't buy 0.3 shares of Berkshire Hathaway ($500k stock)
- Some stocks have high prices

2. Odd lots:

- Buying non-round lots (e.g., 37 shares) can have worse execution

3. Market impact:

- Large orders move prices against you
- Not an issue for small investors
- Major issue for institutional investors

4. Taxes:

- Selling winners triggers capital gains tax
- Might be better to hold suboptimal portfolio than pay 20% tax
- Tax-loss harvesting opportunities

5. Behavioral factors:

- Can you actually hold 80% AAPL through a 50% drawdown?
- Optimal on paper ≠ optimal for your psychology
- Better to hold a "suboptimal" portfolio you can stick with

7.5 When the Model Fails

Scenario 1: One stock dominates (100% allocation)

Why it happens:

- That stock genuinely had best Sharpe ratio in period
- Mathematically correct but practically dangerous

Solutions:

- Add MAX_WEIGHT constraint
- Question the data (is return estimate too optimistic?)
- Use longer time period

Scenario 2: Optimal portfolio has lower Sharpe than individual stock

Why it happens:

- Constraints too restrictive
- Stocks highly correlated (little diversification benefit)
- One stock truly dominates

Solutions:

- Relax constraints
- Add more diverse stocks
- Accept that sometimes diversification doesn't help much

Scenario 3: Optimization fails to converge

Why it happens:

- Constraints are contradictory (e.g., MIN_WEIGHT = 0.15 with 10 stocks = 150% total)
- Numerical issues (very small/large numbers)
- Bad initial guess

Solutions:

- Check constraint logic
- Use different optimizer settings
- Try different initial weights

Scenario 4: Results are unstable

Why it happens:

- Small changes in data → big changes in optimal weights
- Sign of overfitting
- Return estimates have high uncertainty

Solutions:

- Use regularization (penalize extreme weights)
- Use robust optimization methods
- Accept more diversified (suboptimal) portfolio

7.6 The Academic vs Practical Divide

What academics say: "The optimal portfolio maximizes Sharpe ratio subject to constraints."

What practitioners do:

- Start with market-cap weighting (passive index)
- Make tactical tilts based on views
- Use risk budgets (allocate risk across strategies)
- Incorporate transaction costs explicitly
- Use factor models instead of historical returns
- Stress test against multiple scenarios
- Consider client-specific constraints

Why the difference:

- Academics assume perfect information
- Practitioners know returns are unpredictable
- Real money has consequences
- Better to be approximately right than precisely wrong

7.7 Best Practices

1. Use optimization as a starting point, not the final answer

- Does it pass the "smell test"?
- Would you actually invest this way?

2. Add realistic constraints

- MAX_WEIGHT = 0.30 to 0.40 (prevent concentration)
- Consider MIN_WEIGHT if you want forced diversification

3. Use conservative return estimates

- Don't assume 30% returns continue forever
- Adjust historical returns toward long-term averages (10%)
- This is called "shrinkage" in statistics

4. Test on multiple time periods

- Does 2015-2020 give similar results to 2020-2024?
- If wildly different, be skeptical

5. Combine with qualitative judgment

- Do you believe in the company's future?
- Are there industry trends not in historical data?
- Your knowledge matters!

6. Start small

- Don't immediately invest your life savings
- Paper trade first (track performance without real money)
- Gradually increase allocation as you gain confidence

7. Plan for rebalancing

- Set a schedule (quarterly? annually?)
- Set thresholds (rebalance if weight drifts >5%?)
- Consider tax implications

8. Document your process

- Why did you choose these stocks?
- What were the optimal weights?
- What were your assumptions?
- Review periodically - were you right?

Conclusion

Portfolio optimization is a powerful tool that combines:

- **Mathematics:** Rigorous statistical and optimization theory
- **Finance:** Understanding of risk, return, and diversification
- **Programming:** Implementation of complex algorithms
- **Judgment:** Knowing when to trust the model and when to override it

Key takeaways:

1. **Diversification is powerful** - the only free lunch in finance
2. **Correlation matters** - it's not just about individual stock risk
3. **Optimization adds value** - but don't expect miracles
4. **Past performance ≠ future results** - use with caution
5. **Constraints make results practical -** don't let math override common sense

This project demonstrates:

- Modern Portfolio Theory in action
- Real-world data analysis and visualization
- Constrained optimization techniques
- The gap between theory and practice

Next steps for learning:

- Try different stock combinations
- Experiment with different time periods
- Add more sophisticated features (transaction costs, taxes)
- Study when optimization works well vs poorly
- Compare to simple strategies (equal-weight, market-cap weight)

Remember: The goal isn't to find the "perfect" portfolio (it doesn't exist), but to make better-informed investment decisions by understanding risk-return tradeoffs systematically.

Appendix: Key Formulas Reference

Returns:

Daily Return = (Price_t - Price_{t-1}) / Price_{t-1}

Annual Return = Daily Return × 252

Risk:

Variance = σ^2 = Σ(Return_i - Mean)² / n

Volatility = σ = sqrt(Variance)

Annual Volatility = Daily Volatility × sqrt(252)

Covariance:

Cov(A,B) = Σ[(Return_A - Mean_A) × (Return_B - Mean_B)] / n

Correlation:

Corr(A,B) = Cov(A,B) / (σ_A × σ_B)

Portfolio Return:

R_p = Σ(w_i × R_i) = w^T · μ

Portfolio Variance:

σ_p² = w^T · Σ · w

Sharpe Ratio:

Sharpe = (R_p - R_f) / σ_p

Optimization Problem:

Maximize: Sharpe Ratio

Subject to:

- Σw_i = 1 (weights sum to 100%)
- w_i ≥ 0 (no short selling)
- w_i ≤ max_weight (optional)

- $w_i \geq min\_weight$ (optional)

End of Documentation